An experimental understanding of Damping-off disease in red pine seedlings to sustainably control its impact without synthetic herbicides

Abstract

The Wilson State Nursery (WSN) in Boscobel, Wisconsin, supplies millions of red pine seedlings to reforestation efforts throughout the state. Dr. Nick Balster and his lab have monitored soil fertility at WSN for 20 years. In spring of 2021, WSN managers observed the significant fungal (*Fusarium spp.)* disease *damping off* among its two-year-old red pine seedlings. During the summer of 2021, I collected soil samples at WSN in a directed study to measure soil temperature and moisture in relation to this disease. Although those observations yielded some suggestive results, it was clear an experimental study was needed. Thus, I propose here to conduct a controlled greenhouse study to experimentally manipulate soil moisture to examine the degree of infection of red pine seeds that house the pathogenic fungal colonies found at WSN. Little is known about the relationship between *Fusarium spp.* and the environmental factors that facilitate its infection of red pine. This experiment will not only help to fill gaps in the existing literature related to the environmental conditions controlling damping off, but also help elucidate the management practices that limit the impact of this disease and ensure a continual supply of red pines critical to the managed forests of Wisconsin.

Introduction and Literature Review

Wilson State Forest Nursery (WSN) stands as the one and only producer of tree seedlings for reforestation efforts and erosion control in the state of Wisconsin. For example, the nursery played a critical role in repopulating red pine in Northern Wisconsin following a devastating windstorm in July of 2019. The nursery is managed by the Department of Natural Resources (DNR) and is an essential component to the $24.5-billion forest industry of Wisconsin, currently employing over 140,000 jobs (WDNR, n.d.). The nursery also provides full-time and seasonal work for the surrounding community of Boscobel where it is located. Thus, any threat to the seedling output from this nursery would be devastating and thus must be minimized to ensure the sustainability of the forests of Wisconsin and the industry it supports.

Recently, such a threat, a disease called “damping off”, has compromised the health of the nursery resulting in the death of many two-year-old red pine seedlings over the past couple years. Damping off is caused by a fungal pathogen (most commonly *Fusarium spp.*) that attacks the seedling during its germination. Managers typically detect damping off by emerging patches of dead or rotting seedlings. In spring of 2021, damping off was detected at WSN, but a cause, whether management related or

environmental or both, could not be identified; the nursery had sufficient mycorrhizal inoculation and was fumigated, practices used in the past to successfully control this disease (Cram, M. M. et al., 2012). Consequently, nursery managers are keen to understand the environmental factors (e.g., soil conditions) that promote this disease so that management practices can be modified to prevent its spread into other areas of the nursery.

Past studies have identified soil moisture, soil nutrient composition, pH, and temperature as leading environmental factors controlling the vitality of *Fusarium spp.* in red pine. It is important to point out that even though a seed may be infected with *Fusarium spp.*, the abiotic soil factors surrounding the seed might not allow the pathogen to cause terminal harm to the seed (Cram, M.M. et al., 2012; Cruz et al., 2020, Lamichhane et al., 2017). Thus, determining the abiotic components (such as soil moisture) promoting disease spread is essential in developing a treatment plan at WSN.

In the summer of 2021, I conducted a preliminary, observational study comparing the soil conditions of a diseased area of the nursery and an unaffected area to identify possible environmental contenders that caused the sudden spread in damping off. Soil temperature, at the surface and at a 10 cm depth, varied by only a few degrees between the diseased and healthy red pine locations, leading me to conclude that soil temperature was not the primary factor contributing to the damping off incidence, at least during the hottest part of the season. Similarly, two critical measures of soil fertility in the control of this disease, total nitrogen, and pH (Lamichhane et al., 2017), also did not vary between locations. The total nitrogen content for the diseased red pine soil was 0.1 ± 0.02% and the healthy red pine soil was 0.1 ± 0.02%. Likewise, the average pH for the infected soil (5.7 ± 0.22) was similar to the uninfected soil (5.7 ± 0.31).

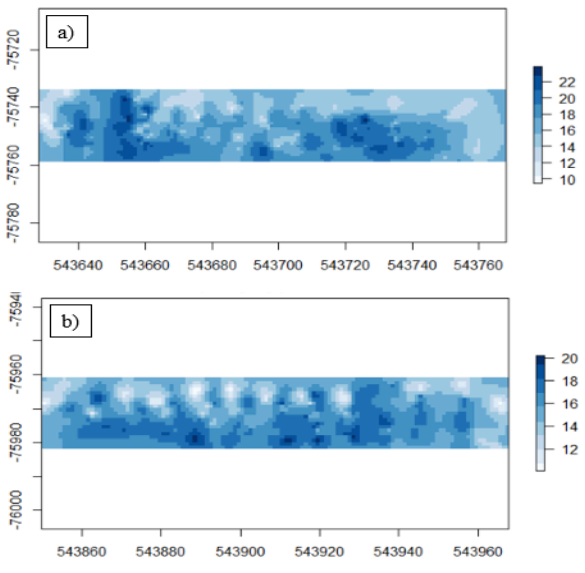


Fig. 1 The volumetric moisture content of (a) diseased red pine and (b) healthy red pine

In contrast to soil fertility and temperature, the preliminary observations of soil moisture suggest that this variable may play a key role in the vitality of this disease at WSN. Low amounts of soil moisture have been observed in other studies to promote the emergence and spread of *Fusarium spp.* (Liu et al., 2020; Cruz et al., 2020). At WSN, I found damping off was present in the areas with lower soil moisture relative to healthy areas (figure 1). Although these observations suggest differences in soil moisture may control the infection and spread, we currently lack experimental data to properly test this hypothesis and draw such an inference.

Therefore, in the proposed study*, I hypothesize that lower soil moisture (experimentally manipulated) will correlate with a higher rate of establishment, growth, and severity of Fusarium spp. on red pine seeds and seedlings. Further, the volumetric moisture content at its permanent wilting point (PWP) will promote greater seedling mortality and reduced height growth compared to seedlings growing above PWP because of Fusarium spp. affinity towards consistently low soil moisture conditions.* This experiment will improve our understanding of the relationship between *Fusarium spp.* and seedling disease and thus help nursery managers control the spread and economic loss of this disease. Moreover, this study nicely combines basic and applied science, as well as innovative methodological techniques, all of which fit with my goals to attend graduate school and combine biological engineering with environmental science to solve real-world problems.

Methods

I will use a greenhouse to test the role of volumetric moisture content (independent variable) on the emergence and severity of damping off (dependent variable). The response variable will be assessed by quantifying the infection rate of red pine seedlings via number of unsuccessfully germinated seedlings or severity of disease post emergence. All seeds that do not germinate, as well as seedlings showing infection will be analyzed to ensure a fungal response.

The experimental design will consist of four experiment runs of 12 separate cylinder pots, each containing 20 red pine seeds purchased from the same source as WSN and soil taken directly from WSN to ensure similar soil texture and structure. The seeds will be planted at a similar density to WSN. Each

pot will be maintained at a different volumetric moisture ranging from ambient moisture to saturation. The volumetric moisture content of each pot will be maintained through drip irrigation because it is easier to manage and maintain compared to a sprinkler system (the irrigation method used by WSN).

Each red pine seed will be inoculated from fungal colonies grown in the lab from the original *Fusarium spp.* strain found at WSN and cultured by the WDNR in late spring of 2021 (and then re-cultured in December of 2021 for research purposes). To inoculate the red pine seeds, they will be rinsed with double distilled water (DDW), soaked in NaOCl 1% for 1 min, and rinsed again. Next, the seeds will be placed in petri dishes of potato dextrose agar and 6-mm-diameter disc from the *Fusarium spp.* colony (Kalman et al., 2020). After inoculation, the seeds will be incubated for two to three days at 28 ± 1° C in the dark. If needed, re-culturing of the fungi for future experiment runs will be performed by plating newly infected red pine tissue (from previous experiment runs) in separate petri dishes. I will be working closely with a WDNR plant pathologist to ensure the correct culturing and inoculation procedures are used. After the fungus invades the seeds, they will be planted in the 12 cylinder pots divided equally among the moisture and temperature treatments.

Four different moisture levels (treatments) will be maintained in this experiment: no moisture (control), wilting point volumetric moisture (-1.5 Mpa), field capacity volumetric moisture (-0.33 Mpa), and saturated soil moisture (0 Mpa). The greenhouse temperature will be measured throughout the experiment to quantify any interaction between soil moisture and temperature, however we will work to keep the greenhouse temperature between 16°C and 30°C (ideal temperature range for germination) (USDA, n.d.). Each pot will be outfitted with METER Inc. TM-5© soil moisture and temperature sensors linked to data loggers for consistent data collection throughout the experiment. Measurements will be automatically taken at hourly intervals and then averaged every twelve hours throughout the duration of the study. Finally, morphological measurements including color, height, and overall vitality (subjective) of each seedling on a weekly basis.

Timeline for Experiment

Month	Plan
February-March	Set up greenhouse; purchase the uninfected red pine seeds; calibrate the METER Inc. soil probes©; build the drip irrigation system; collect soil from WSN; warm the soil to optimum red pine germination temperature.
April-May	Complete the Fusarium spp. colony culturing and subsequent inoculation of the number of seeds needed for the first experiment.
June-September	Four total experiment runs; fifth experiment run if needed.

Conclusion and Future Direction

The results of this study will improve our understanding of the relationships among specific environmental conditions and damping off in red pine seedlings to better control the economic and agricultural loss from this pathogen without the use of herbicides. This experiment will allow me to learn and practice the design, management, and interpretation of a controlled and publishable experiment that not only expands our knowledge about a plant disease but will also result in management recommendations for nurseries suffering from this issue. Finally, this experience will also allow me to pursue my research interests in preparation for my dream of a graduate position that combines biological systems engineering and soil science.

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